

- 2 (a) The impulse of a force acting on a particle defined as the time-integral of the force with respect to time.

(b) (i) Impulse = area under the force-time graph

$$= \frac{1}{2}(0.75 - 0.25)(-0.32) = -0.080 \text{ N s}$$

Impulse = change in momentum

$$m\Delta v = -0.080 \text{ N s}$$

$$\Delta v = \frac{-0.080}{0.150} = -0.533 \text{ m s}^{-1}$$

Hence, magnitude of change of velocity = 0.53 m s^{-1}

- (ii) 1. The velocity of the mass decreases at an increasing rate from 0.267 m s^{-1} until it comes to a rest at $t = 0.50 \text{ s}$.

2. The velocity of the mass increases from rest at a decreasing rate until it reaches 0.267 m s^{-1} in the opposite direction.

3 (a) (i) The direction of the resultant force acting on the object must be perpendicular to